Compound Cyclone Exposure

Is Time Since Last Strike Randomly Assigned?

Section 1

The Baseline Identification

Step 1: What Is Plausibly Randomly Assigned?

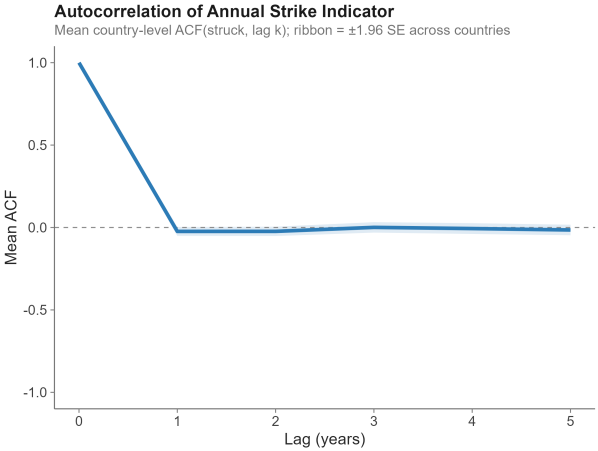
Claim: conditional on a country's long-run climatology $\bar{W}_i$, the realised wind speed in any given year is approximately exogenous.

- **Why?** TC track and intensity are governed by sea-surface temperatures, wind shear, and stochastic convective dynamics — not by a country's GDP path.
- **Formally:** $w_{it} \perp \varepsilon_{it} \mid \bar{W}_i,\ \alpha_i,\ \delta_{r(i)t}$ where $\alpha_i(t)$ is a country trend and $\delta_{r(i)t}$ is a region$\times$year FE.
- **Evidence:** ACF of the binary strike indicator is near zero at all lags $\Rightarrow$ strikes are approximately i.i.d. within country from year to year.

What this supports: a within-country LP estimating the GDP response to an unexpected wind shock, evaluated as a function of $\bar{W}_i$.

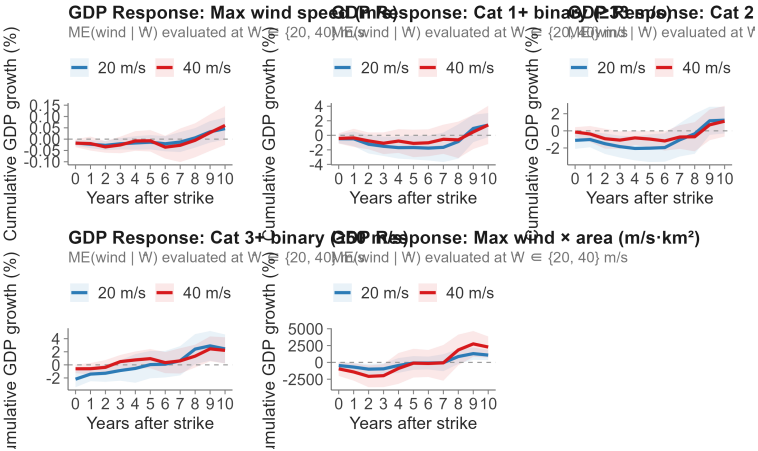
What this does *not* support: identification of how the response depends on *when* the country was last struck — that requires a separate argument.

Autocorrelation of Annual Strike Indicator



Mean country-level ACF of $\mathbf{1}[\text{maxwind} > 0]$, pooled across all countries. Ribbon $= \pm 1.96\,\text{SE}$. Near-zero ACF at all lags supports the i.i.d.-within-country assumption underlying the LP.

Baseline LP: GDP Response by Country Climatology



State-dependent LP at $\bar{W} \in \{20, 40\}$ m/s, five shock definitions. Blue = low-climatology ($\bar{W} = 20$), red = high-climatology ($\bar{W} = 40$). This motivates asking whether *recent prior exposure* moderates the response.

Section 2

The Compound Question and Its Identification Problem

The Compound Question

Question: does the GDP cost of a cyclone depend on how recently the country was last struck?

Two competing stories:

Mechanism	Prediction
Compounding damage (fiscal space exhausted, infrastructure already damaged)	GDP hit is *larger* when the prior strike was recent
Adaptation / reduced stock (less capital to destroy, households diversified)	GDP hit is *smaller* when the prior strike was recent

Approach: estimate $\text{ME}(w_{it} \mid \tau_{it} = c) = \hat{\beta}_1 + \hat{\beta}_2 \cdot c$ where τ_{it} = years since the country's previous TC event.

The identification problem: τ_{it} is not randomly assigned.

Why τ Is Not Randomly Assigned

Mechanical link: a country with high average wind climatology $\bar{W}_i$ is struck *more often* $\Rightarrow$ it has shorter inter-strike gaps τ on average — by construction.

Formally, the state variable τ_{it} is a function of past shocks:

$$\tau_{it} = t - \max\{s < t : w_{is} > 0\}$$

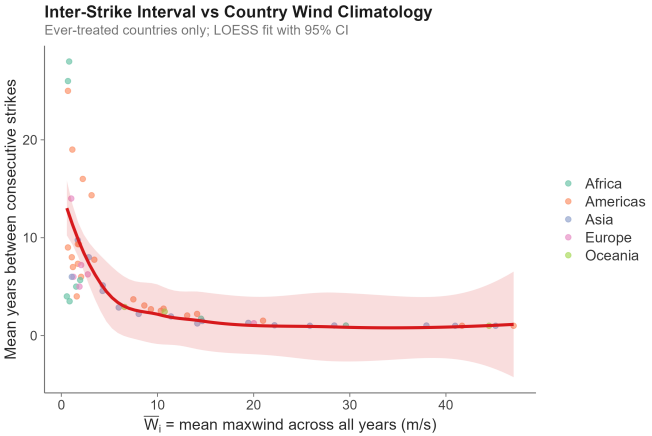
Its distribution depends on the Bernoulli success probability $p_i = P(\text{struck}_t)$, which is a monotone function of $\bar{W}_i$.

Consequence: if we find that the GDP response is larger when τ is large (isolated event), we cannot immediately conclude this is a *time-since-last-strike* effect. It could instead be:

- A **cross-country adaptation effect**: isolated events happen disproportionately in low-$\bar{W}$ countries, which are less adapted.
- A **selection effect**: never-struck periods are more common for rarely-struck countries.

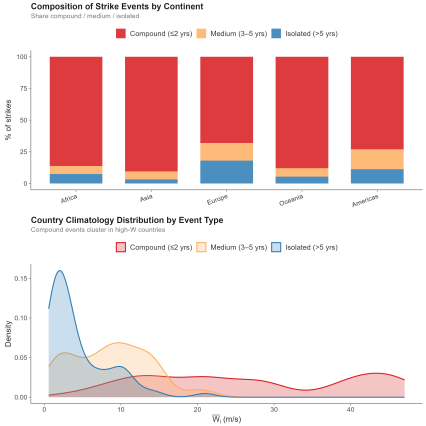
$\Rightarrow$ We need to establish this empirically and then design robustness tests.

EDA: High-Climatology Countries Have Shorter Inter-Strike Gaps



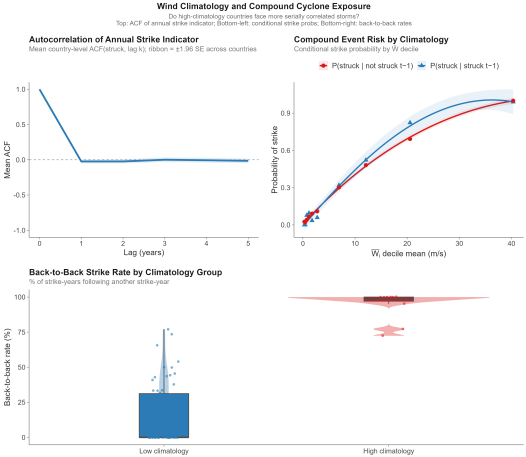
Ever-treated countries ($\bar{W}_i > 0$). X: $\bar{W}_i$ (m/s). Y: mean years between strikes. LOESS fit $\pm 95\,\%$ CI. Higher $\bar{W} \Rightarrow$ shorter gaps $\Rightarrow$ lower τ. This is mechanical, not behavioural.

EDA: Compound Events Cluster in High-Climatology Countries



Top: share of compound ($\tau \leq 2$), medium ($\tau \in [3, 5]$), and isolated ($\tau > 5$) events by continent. **Bottom:** $\bar{W}_i$ distribution by event type — compound events cluster in high-climatology countries. τ and $\bar{W}$ are strongly correlated: any compound-LP gradient partially reflects the climatology gradient.

Back-to-Back Strikes: Rate by Climatology Decile



Left: $P(\text{struck}_t \mid \text{struck}_{t-1})$ vs. $P(\text{struck}_t \mid \text{not struck}_{t-1})$ by $\bar{W}$ decile. **Right:** back-to-back strike rate by climatology group. High-$\bar{W}$ countries have substantially elevated compound probabilities — τ is not randomly assigned.

Identification Summary

Object	Randomly assigned?	Identification lever
w_{it} (maxwind in year t)	**Yes**, conditional on $\bar{W}_i$ and FEs	Within-country year-to-year variation
$\bar{W}_i$ (long-run climatology)	No — time-invariant geography	Cross-country; absorbed by FE as level
τ_{it} (years since last TC)	**No** — determined by history of w_{is}; systematically shorter for high-$\bar{W}_i$ countries	Cannot be isolated from $\bar{W}_i$ without further assumptions

The compound LP estimates $\beta_1 + \beta_2 \cdot \tau$ but β_2 conflates the *time-since-last-event* mechanism with the cross-sectional *adaptation-to-climatology* mechanism. Robustness tests (within-climatology splits, LOCO, ever-treated) probe the severity of this confound; a valid IV would solve it.

Section 3

Compound LP: Specification and Methods

Compound LP: The Model

For each horizon $h = 0, \ldots, 10$, estimate by OLS:

$$\log \mathrm{GDP}_{i,t+h} - \log \mathrm{GDP}_{i,t-1} = \beta_1^{(h)} w_{it} + \beta_2^{(h)}(w_{it} \times \tau_{it}) + \boldsymbol{\gamma}^{(h)\prime} X_{it} + \alpha_i^{(h)}(t) + \delta_{r(i)t}^{(h)} + \varepsilon_{iht}$$

Symbol	Definition
w_{it}	Shock variable (five variants)
τ_{it}	Years since last maxwind > 0 (continuous state variable)
X_{it}	$\ell(w_{it}, 1{:}2)$ and $\ell(\Delta \log \mathrm{GDP}_{it}, 1{:}2)$ (two lags of shock and GDP growth)
$\alpha_i^{(h)}(t)$	Country quadratic trend (`countrycode[year]` + `countrycode[year²]`)
$\delta_{r(i)t}^{(h)}$	Region×year FE (`region^year`)

Marginal effect at $\tau = c$ (delta-method SE):

$$\widehat{\mathrm{ME}}(c) = \hat{\beta}_1^{(h)} + \hat{\beta}_2^{(h)} \cdot c, \quad \widehat{\mathrm{SE}}(c) = \sqrt{\hat{V}_{11} + c^2 \hat{V}_{22} + 2c\, \hat{V}_{12}}$$

Evaluated at $c \in \{1, 3, 5\}$ years.

Methods: State Variable, Residualisation, and SE

1. State variable (`time_since_last`): $\tau_{it} = t - \max\{s < t : w_{is} > 0\}$; `NA` before the first ever strike. At the strike year, τ records the gap since the *previous* strike, not zero.

2. Residualised shock: $\tilde{w}_{it} = \text{maxwind}_{it} - \hat{\mu}_{it}$ where $\hat{\mu}_{it}$ from `feols(maxwind ~ 1 | countrycode[year] + region^year)`. Removes country linear trends and region×year patterns; retains idiosyncratic wind variation (cor ≈ 0.52).

3. SE: Driscoll–Kraay (`vcov = DK ~ year`) — arbitrary serial correlation within country and spatial correlation across countries in the same year.

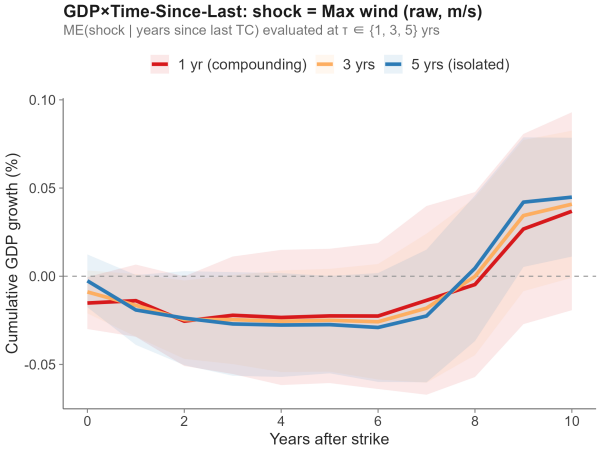
4. Five shock variables, all interacted with τ^{any}:

Name	R variable	Threshold
Raw maxwind	`maxwind`	continuous (m/s)
Residualised maxwind	`maxwind_resid`	net of trends/FEs
Cat 1+ binary	`wind_cat1plus`	≥33 m/s
Cat 2+ binary	`wind_cat2plus`	≥43 m/s
Cat 3+ binary	`wind_cat3plus`	≥50 m/s

Section 4

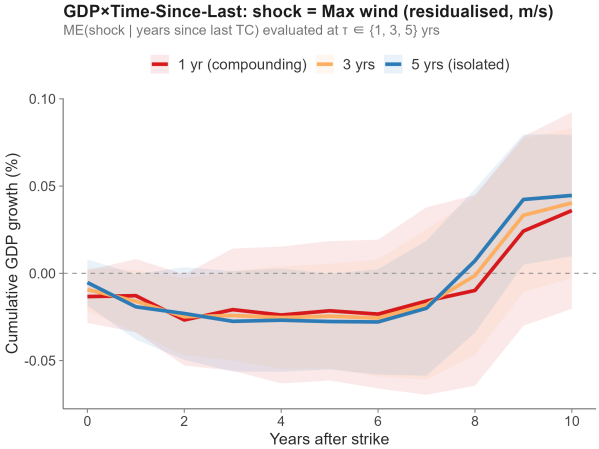
Results: IRFs Across Five Shock Measures

IRF: Raw Maxwind × Time Since Last TC



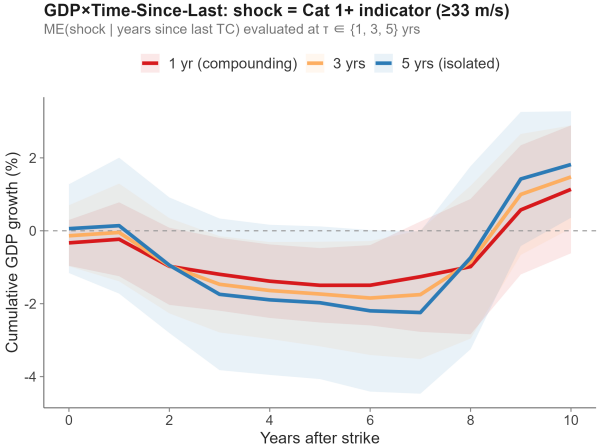
Shock = raw max wind speed (m/s). Red = compounding ($\tau = 1$ yr), gold = medium gap ($\tau = 3$ yrs), blue = isolated ($\tau = 5$ yrs). Gradient: isolated events generate slightly larger negative GDP effects, consistent with both the adaptation and the $\bar{W}$ confound stories.

IRF: Residualised Maxwind × Time Since Last TC



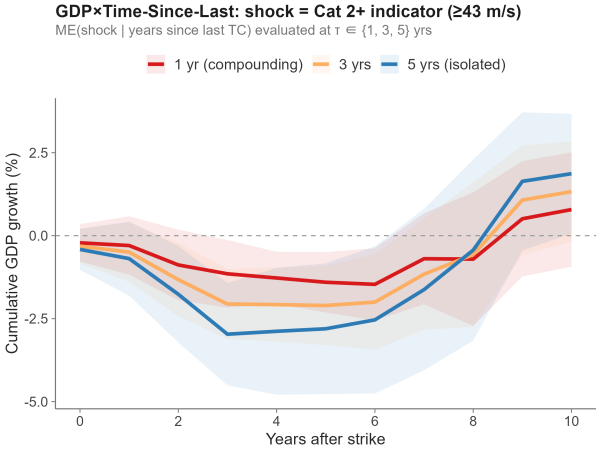
Shock = $\tilde{w}_{it}$ (maxwind net of country linear trend and region×year FE; cor ≈ 0.52 with raw maxwind). Residualisation isolates idiosyncratic within-country wind variation. Gradient largely unchanged, suggesting the confound is not driven by predictable cyclone-season patterns.

IRF: Cat 1+ Binary × Time Since Last TC



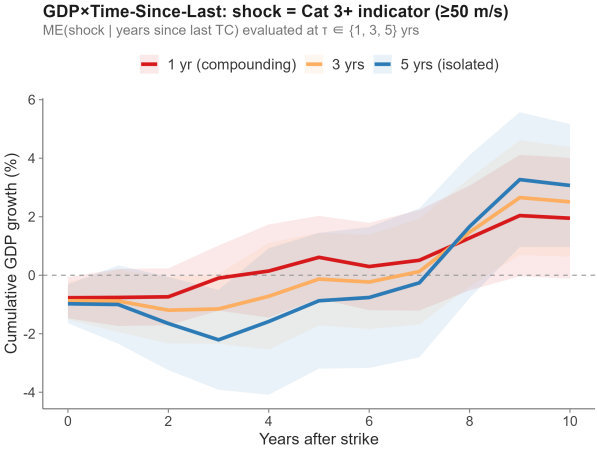
Shock = Cat 1+ indicator ($\geq$33 m/s). Gradient is clear: isolated events ($\tau = 5$, blue) generate larger GDP losses than compounding events ($\tau = 1$, red).

IRF: Cat 2+ Binary × Time Since Last TC



Shock = Cat 2+ indicator ($\geq$43 m/s). The gradient sharpens: at $h = 5$, the isolated-minus-compounding gap is larger than for Cat 1+, suggesting the mechanism operates more strongly for intense events.

IRF: Cat 3+ Binary × Time Since Last TC



Shock = Cat 3+ indicator ($\geq$50 m/s). Fewer treated observations; wider confidence bands. Back-to-back major events ($\tau = 1$) show attenuated GDP cost — consistent with reduced destructible stock after a prior major event.

Section 5

Robustness: Probing the $\bar{W}$ Confound

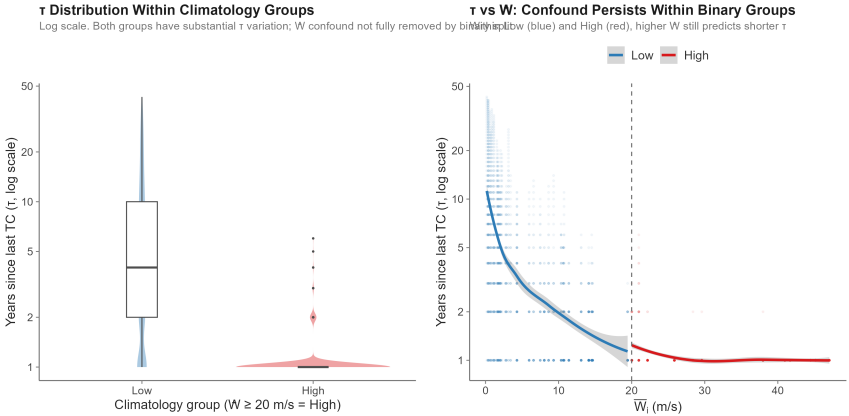
Strategy: Three Tests for $\bar{W}$ Confounding

The gradient (isolated > compounding GDP loss) could reflect:

1. **Dilution by never-struck controls:** never-struck countries have $\tau = \texttt{NA}$ and are dropped from the compound LP, but their presence in the full panel may affect estimates. *Fix:* restrict to ever-treated countries ($\bar{W}_i > 0$).
2. **Geographic concentration:** compound events cluster in specific regions (East/Southeast Asia, Caribbean) with distinct recovery patterns. *Fix:* leave-one-continent-out (LOCO).
3. $\bar{W}$ **composition:** the gradient is entirely explained by comparing high-$\bar{W}$ countries (short τ) to low-$\bar{W}$ countries (long τ). *Fix:* estimate the gradient *within* high- and low-climatology groups separately. If it vanishes within groups, τ has no independent variation.

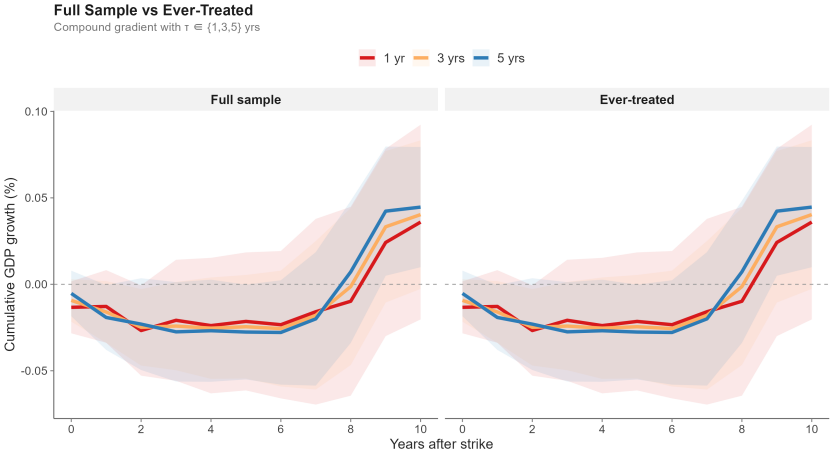
Spoiler: Test 3 fails for a data reason — τ has essentially *no variation* within the High-$\bar{W}$ group (98 % of obs have $\tau \leq 2$). And the residualised-τ LP (Test 4 below) shows the gradient disappears once the $\bar{W}$–τ correlation is removed.

Diagnostic: τ Has Almost No Variation in High-$\bar{W}$ Countries



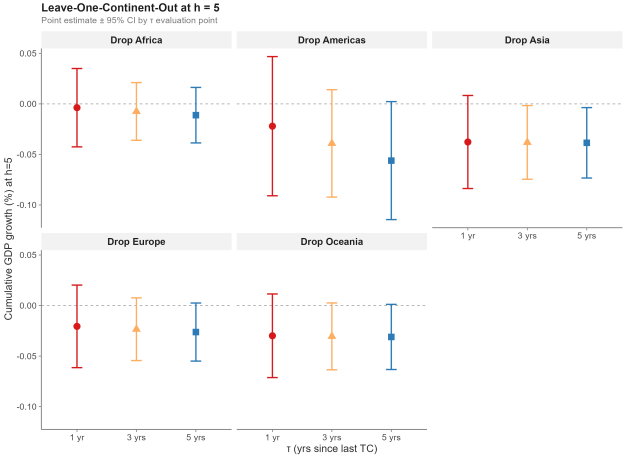
Left: τ distribution (log scale) within binary $\bar{W}$ groups. **Right:** within each group, higher $\bar{W}$ still predicts shorter τ (LOESS). **Key fact:** in the High group ($\bar{W} \geq 20$), 98.5 % of obs have $\tau \leq 2$ yrs — effectively *no variation* in τ. Any "gradient" estimated within High is extrapolation, not identification.

Full Sample vs Ever-Treated



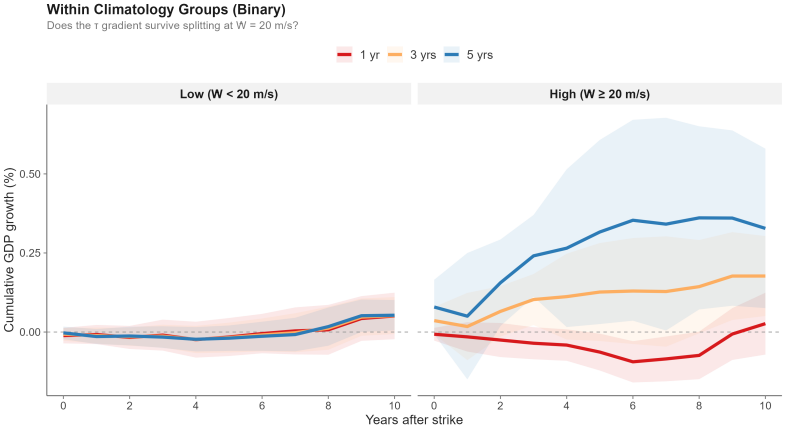
Left: full panel. Right: ever-struck countries only ($\bar{W}_i > 0$). Gradient is similar in both samples — not driven by never-struck controls.

Leave-One-Continent-Out



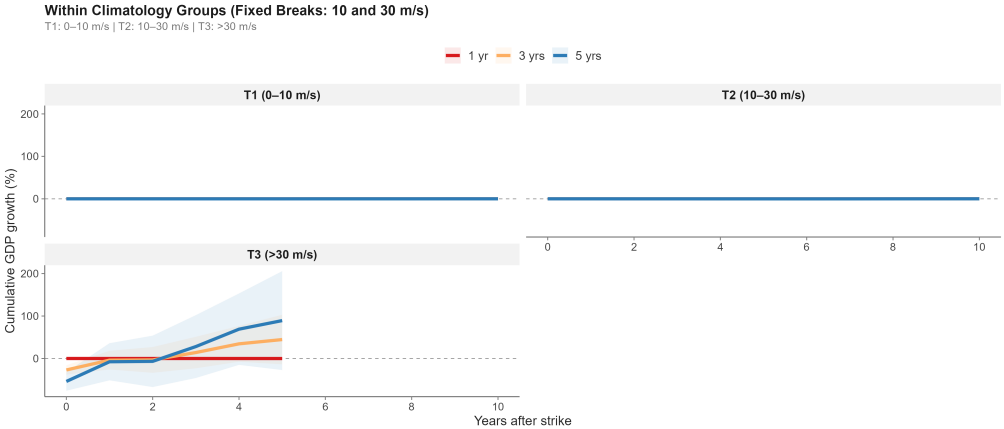
Maxwind-resid IRF at $h = 5$, dropping one continent at a time. No single region drives the pattern.

Within Binary and Tercile $\bar{W}$ Groups



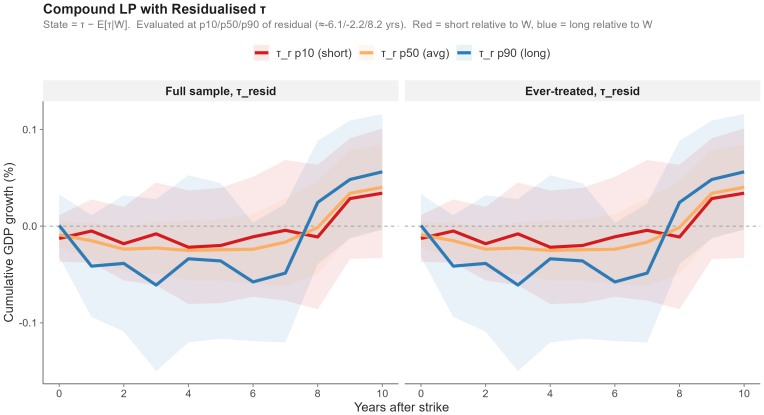
Binary split at $\bar{W} = 20\,\mathrm{m/s}$. In the **Low** group (blue), the gradient is flat — $\tau = 1, 3, 5$ lines overlap. In the **High** group (red), a large apparent gradient emerges, but this is an artefact: 98.5 % of High-group obs have $\tau \leq 2$, so the LP extrapolates to $\tau = 5$ with virtually no support.

Within $\bar{W}$ Groups (Fixed Breaks: 10 and 30 m/s)



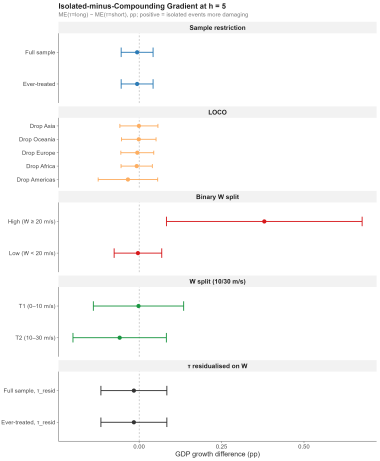
T1 (0–10 m/s, τ mean = 9 yrs, 58 % isolated): gradient flat, near zero. T2 (10–30 m/s, τ mean = 1.6 yrs, 85 % compound): minimal τ variation, gradient poorly identified. T3 (>30 m/s): 100 % $\tau \leq 2$ — LP extrapolates wildly ($\widehat{\text{gradient}} = 89$ pp), excluded from summary.

Key Test: Residualise τ on $\bar{W}$



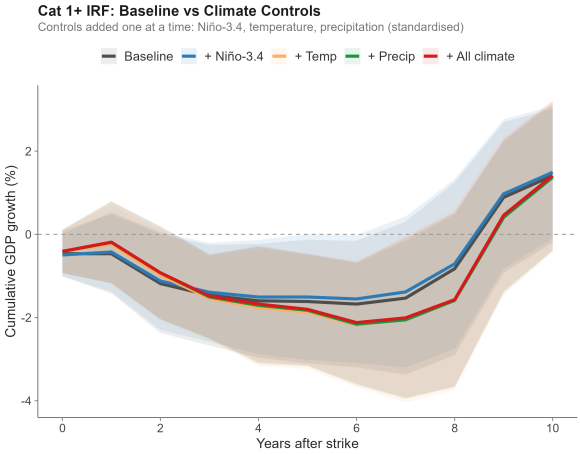
State variable $= \hat{\tau}_{\text{resid}} = \tau_{it} - \hat{E}[\tau_{it} \mid \bar{W}_i]$ (OLS projection; $\text{cor}(\bar{W}, \hat{\tau}_{\text{resid}}) = 0$ by construction). Evaluated at p10 / p50 / p90 of residual (≈ -6, -2, $+8$ yrs). **The gradient is indistinguishable from zero** (-2 pp, CI: $[-12, +8]$). Removing the $\bar{W}$–τ correlation kills the compound effect.

Gradient Summary: the Compound Effect Is a $\bar{W}$ Confound



ME(long τ) − ME(short τ) at $h = 5$. Positive in full-sample and LOCO tests; collapses to zero in T1 and T2 (10/30 m/s breaks) and in the residualised-τ LP. T3 (>30 m/s) excluded: 100 % $\tau \leq 2$, LP extrapolates. The compound gradient is **a $\bar{W}$–τ confound**.

ENSO and Climate Controls



Cat 1+ baseline LP vs adding Niño-3.4, temperature, and precipitation anomalies (standardised) as contemporaneous controls. GDP effect is stable across specifications $\Rightarrow$ omitted climate variables do not explain the cyclone–GDP response, strengthening the causal interpretation of β_1.

Section 6

Remedies: Towards Valid Identification of τ

What Would a Valid Instrument for τ Look Like?

Goal: instrument Z_{it} satisfying (i) **relevance**: shifts τ_{it}, and (ii) **exclusion**: affects GDP only via τ_{it}.

Instrument	Relevance	Exclusion concern
Basin-wide TC count in $t-1$ (excl. own country)	High-activity year $\Rightarrow \tau \downarrow$	SST co-moves with agriculture
ENSO state at $t-1$ (La Niña $\Rightarrow$ more Atlantic TCs)	Shifts prior-year strike probability	ENSO directly affects rainfall and output
Lagged strike in adjacent country	Regional TC systems hit neighbours in sequence	Common regional shocks

None of the candidates is clean. The most promising direction is a **shift-share** design: basin-level TC frequency (the "shift") $\times$ country geographic exposure share (time-invariant "share"). The share is predetermined and the shift is driven by large-scale atmospheric dynamics that are plausibly orthogonal to any individual country's GDP path.

Practical Partial Remedies (Without a Full IV)

1. **Within-group estimation:** narrow the $\bar{W}$ bins (terciles, quartiles) to minimise the within-cell τ–$\bar{W}$ correlation. Residual variation in τ within fine bins is closer to exogenous.
2. **Residualise τ on $\bar{W}$:** project τ_{it} on $\bar{W}_i$ (or a flexible function) and use the residual as the state variable. Removes cross-sectional correlation while retaining within-country variation in inter-strike timing.
3. **Matched sample:** match high-τ and low-τ observations on $\bar{W}$ and income (CEM). Compare within matched pairs.
4. **Bounds:** derive worst-case bounds on β_2 under the assumption that all τ confounding operates through $\bar{W}$ (Manski-type partial identification).

Given the results below, the first priority is the residualised-τ approach: it is implementable, already estimated, and shows the compound gradient is a $\bar{W}$ confound.

Section 7

Takeaways

Summary

1. **Maxwind is plausibly quasi-random conditional on $\bar{W}$.** Near-zero ACF supports the i.i.d. assumption; ENSO controls do not attenuate the GDP effect, supporting causal interpretation of β_1.
2. **τ is not randomly assigned.** High-$\bar{W}$ countries are struck more often $\Rightarrow$ shorter τ by construction. Within the High group ($W^- \geq 20$), 98.5 % of observations have $\tau \leq 2$ — τ is essentially a constant there.
3. **The apparent compound gradient is a $\bar{W}$ confound.** Within-$\bar{W}$ terciles T1 and T2, the gradient is zero. Once τ is residualised on $\bar{W}$ (removing the cross-country level correlation), the gradient collapses to -2 pp and is statistically indistinguishable from zero (CI: $[-12, +8]$).
4. **"Within climatology groups" is not a valid fix.** The High group has near-degenerate τ variation, so the LP extrapolates rather than identifies. The apparent gradient in High is an artefact of that extrapolation.
5. **Conclusion:** no evidence that time since last TC causally moderates the GDP response once the $\bar{W}$–τ mechanical link is controlled for. A valid IV (shift-share: basin TC count $\times$ country exposure share) would provide a definitive test.